

Лабораторная работа № 5. Эмуляция и измерение потерь пакетов в глобальных сетях

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Цель работы

Основной целью работы является получение навыков проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных. Эти параметры влияют на производительность протоколов и сетей

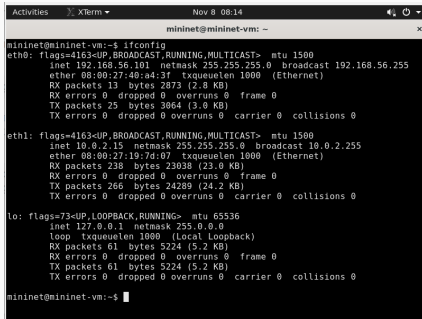
Задание

Задание

1. Задайте простейшую топологию, состоящую из двух хостов и коммутатора с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.
3. Реализуйте воспроизводимый эксперимент по добавлению правила отбрасывания пакетов в эмулируемой глобальной сети. На экран выведите сводную информацию о потерянных пакетах.
4. Самостоятельно реализуйте воспроизводимые эксперименты по исследованию параметров сети, связанных с потерей, изменением порядка и повреждением пакетов при передаче данных. На экран выведите сводную информацию о потерянных пакетах.

Выполнение лабораторной работы

Запуск лабораторной топологии



```
Activities XTerm Nov 8 08:14
mininet@mininet-vm: ~
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.101 netmask 255.255.255.0 broadcast 192.168.56.255
    ether 08:00:27:40:a4:3f txqueuelen 1000 (Ethernet)
    RX packets 13 bytes 2873 (2.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 25 bytes 3064 (3.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    ether 08:00:27:19:7d:07 txqueuelen 1000 (Ethernet)
    RX packets 238 bytes 23038 (23.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 266 bytes 24289 (24.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 61 bytes 5224 (5.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 61 bytes 5224 (5.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet@mininet-vm:~$
```

Рис. 1: Запуск виртуальной машины

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 f234a838fbf82b5ab19eb061bf664dia
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth add mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 f234a838f
bf82b5ab19eb061bf664dia
root@mininet-vm:~# logout
mininet@mininet-vm:~$ xterm
```

Рис. 2: Исправление прав запуска X-соединения

Запуск лабораторной топологии

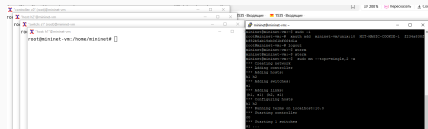


Рис. 3: Задание простейшей топологии

Запуск лабораторной топологии

```
root@mininet-vm:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 6a:1d:55:e2:06:54 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1192 bytes 285088 (285.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1192 bytes 285088 (285.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

Рис. 4: Информация на хосте h1

Запуск лабораторной топологии

```
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.0.2  netmask 255.0.0.0  broadcast 10.255.255.255
    ether fe:66:2a:79:b1:1e  txqueuelen 1000  (Ethernet)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 1474  bytes 284388 (284.3 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 1474  bytes 284388 (284.3 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet#
```

Рис. 5: Информация на хосте h2

Запуск лабораторной топологии

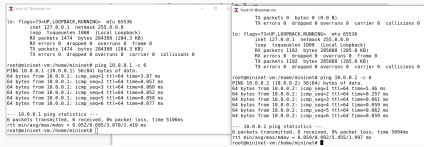


Рис. 6: Проверка подключения между хостами

Минимальное RTT: 0.052 мс, среднее RTT: 0.095 мс, максимальное RTT: 3.840 мс, стандартное отклонение RTT: 1.419 мс для 10.0.0.1. Для 10.0.0.2: минимальное RTT: 0.059 мс, среднее RTT: 0.992 мс, максимальное RTT: 5.435 мс, стандартное отклонение RTT: 1.997 мс. Потерь данных в обоих случаях нет (0% потерь пакетов).

Пакеты могут быть потеряны в процессе передачи из-за таких факторов, как битовые ошибки и перегрузка сети. Скорость потери данных часто измеряется как процентная доля потерянных пакетов по отношению к количеству отправленных пакетов.

```
root@mininet-virtual-machine:~# sudo tc qdisc add dev h1-eth0 root netem loss 10  
%  
root@mininet-virtual-machine:~#
```

Рис. 7: Добавление 10% потерь пакетов

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 100
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.46 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.531 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.160 ms
█
```

Рис. 8: Проверка соединения

```
root@mininet-vn:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem loss 10
root@mininet-vn:/home/mininet#
```

Рис. 9: Добавление 10% потерь пакетов на хосте h2

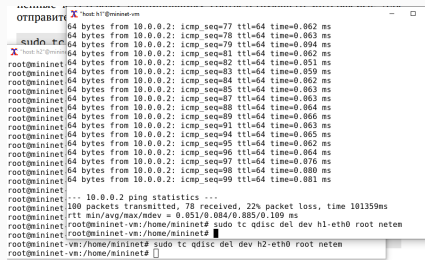
Интерактивные эксперименты

```
host h1" @mininet-vm
64 bytes from 10.0.0.2: icmp_seq=75 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=77 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=78 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=79 ttl=64 time=0.094 ms
64 bytes from 10.0.0.2: icmp_seq=81 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=82 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=83 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=84 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=85 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=87 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=88 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=89 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=91 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=94 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=95 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=96 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=97 ttl=64 time=0.076 ms
64 bytes from 10.0.0.2: icmp_seq=98 ttl=64 time=0.080 ms
64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.081 ms

--- 10.0.0.2 ping statistics ---
100 packets transmitted, 78 received, 22% packet loss, time 101359ms
rtt min/avg/max/mdev = 0.051/0.084/0.885/0.109 ms
root@mininet-vm: /home/mininet#
```

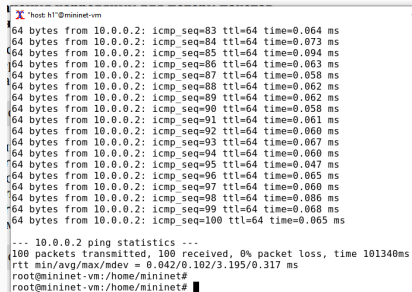
Рис. 10: Проверка, что соединение имеет большой процент потерь

22% потерь пакетов



```
root@mininet:~# host h1@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=77 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=78 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=79 ttl=64 time=0.094 ms
root@mininet:~# host h2@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=81 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=82 ttl=64 time=0.051 ms
root@mininet:~# host h3@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=83 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=84 ttl=64 time=0.062 ms
root@mininet:~# host h4@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=85 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=87 ttl=64 time=0.063 ms
root@mininet:~# host h5@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=88 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=89 ttl=64 time=0.066 ms
root@mininet:~# host h6@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=91 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=94 ttl=64 time=0.065 ms
root@mininet:~# host h7@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=95 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=96 ttl=64 time=0.064 ms
root@mininet:~# host h8@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=97 ttl=64 time=0.076 ms
64 bytes from 10.0.0.2: icmp_seq=98 ttl=64 time=0.080 ms
root@mininet:~# host h9@mininet-vm
отправит 64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.081 ms
root@mininet:~# ... 10.0.0.2 ping statistics ...
root@mininet:~# 100 packets transmitted, 78 received, 22% packet loss, time 101359ms
root@mininet:~# rtt min/avg/max/ndev = 0.051/0.084/0.885/0.109 ms
root@mininet:~# root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet#
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h2-eth0 root netem
root@mininet-vm:/home/mininet#
```

Рис. 11: Восстановление конфигураций



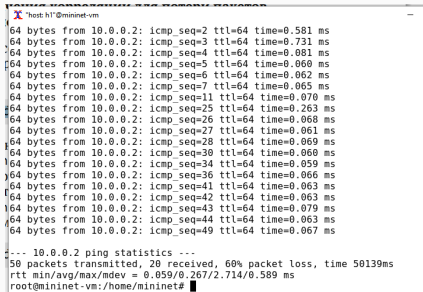
```
host: h1" @mininet-vm
64 bytes from 10.0.0.2: icmp_seq=83 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=84 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=85 ttl=64 time=0.094 ms
64 bytes from 10.0.0.2: icmp_seq=86 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=87 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=88 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=89 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=90 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=91 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=92 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=93 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=94 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=95 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=96 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=97 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=98 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.068 ms
64 bytes from 10.0.0.2: icmp_seq=100 ttl=64 time=0.065 ms

--- 10.0.0.2 ping statistics ---
100 packets transmitted, 100 received, 0% packet loss, time 101340ms
rtt min/avg/max/mdev = 0.042/0.102/3.195/0.317 ms
root@mininet-vm:/home/mininet#
root@mininet-vm:/home/mininet#
```

Рис. 12: Проверка, что соединение не имеет явной потери

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 50  
% 50%  
root@mininet-vm:/home/mininet# █
```

Рис. 13: Добавление коэффициента потери пакетов 50%



```
host: h1@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.581 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.731 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.070 ms
64 bytes from 10.0.0.2: icmp_seq=25 ttl=64 time=0.263 ms
64 bytes from 10.0.0.2: icmp_seq=26 ttl=64 time=0.068 ms
64 bytes from 10.0.0.2: icmp_seq=27 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=28 ttl=64 time=0.069 ms
64 bytes from 10.0.0.2: icmp_seq=30 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=34 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=36 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=41 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=42 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=43 ttl=64 time=0.079 ms
64 bytes from 10.0.0.2: icmp_seq=44 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=49 ttl=64 time=0.067 ms

--- 10.0.0.2 ping statistics ---
50 packets transmitted, 20 received, 60% packet loss, time 50139ms
rtt min/avg/max/mdev = 0.059/0.267/2.714/0.589 ms
root@mininet-vm:/home/mininet#
```

Рис. 14: Проверка, что имеются потери пакетов

60% потеря пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 10
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.86 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.515 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.189 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.057 ms
--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9189ms
rtt min/avg/max/mdev = 0.057/0.399/2.860/0.831 ms
root@mininet-vm:/home/mininet#
```

Рис. 15: Восстановление для узла h1 конфигурации по умолчанию и проверка

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem corrupt  
0.01%  
root@mininet-vm:/home/mininet#
```

Рис. 16: Добавление повреждения пакетов 0,01%

Интерактивные эксперименты

```
root@kali:~/Documents# ifconfig eth0:ens33=9.8.8.10/24 9.8.8.10 ens33
root@kali:~/Documents# sudo ifconfig eth0 dev v1-eth0 root routeen carge
8.0%
root@kali:~/Documents# ifconfig ens33 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ ] local 10.0.0.2 port 4012 connected to 10.0.0.2 port 5201
ID Interval Transfer Bitrate Retx Cong
[ ] 0.00-1.00 sec 923 Kbytes 7.33 Gbits/sec 0 8.10 Mbytes
[ ] 1.00-2.00 sec 1.13 Gbytes 9.87 Gbits/sec 3 2.85 Mbytes
[ ] 2.00-3.00 sec 1.43 Gbytes 12.4 Gbits/sec 2 1.78 Mbytes
[ ] 3.00-4.00 sec 1.43 Gbytes 13.9 Gbits/sec 1 1.35 Mbytes
[ ] 4.00-5.00 sec 1.37 Gbytes 13.5 Gbits/sec 2 365 Kbytes
[ ] 5.00-6.00 sec 1.68 Gbytes 13.7 Gbits/sec 3 815 Kbytes
[ ] 6.00-7.00 sec 1.54 Gbytes 13.3 Gbits/sec 0 1.70 Mbytes
[ ] 7.00-8.00 sec 1.63 Gbytes 13.6 Gbits/sec 4 371 Kbytes
[ ] 8.00-9.00 sec 1.43 Gbytes 12.6 Gbits/sec 2 324 Kbytes
[ ] 9.00-10.00 sec 1.58 Gbytes 12.7 Gbits/sec 2 1.58 Mbytes
- - - - -
ID Interval Transfer Bitrate Retx Cong
[ ] 0.00-10.00 sec 10.4 Gbytes 12.4 Gbits/sec 39 sender
[ ] 0.00-10.01 sec 10.4 Gbytes 12.4 Gbits/sec 39 receiver
ipert Demo
root@kali:~/Documents#
```

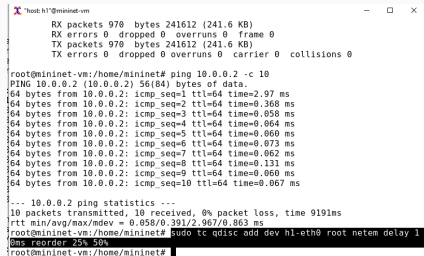
```
root@kali:~/Documents# sudo ifconfig eth0 dev v1-eth0 root routeen
root@kali:~/Documents# ifconfig ens33 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ ] local 10.0.0.2 port 4012 connected to 10.0.0.2 port 5201
ID Interval Transfer Bitrate Retx Cong
[ ] 0.00-1.00 sec 923 Kbytes 7.33 Gbits/sec 0 8.10 Mbytes
[ ] 1.00-2.00 sec 1.13 Gbytes 9.87 Gbits/sec 3 2.85 Mbytes
[ ] 2.00-3.00 sec 1.43 Gbytes 12.4 Gbits/sec 2 1.78 Mbytes
[ ] 3.00-4.00 sec 1.43 Gbytes 13.9 Gbits/sec 1 1.35 Mbytes
[ ] 4.00-5.00 sec 1.37 Gbytes 13.5 Gbits/sec 2 365 Kbytes
[ ] 5.00-6.00 sec 1.68 Gbytes 13.7 Gbits/sec 3 815 Kbytes
[ ] 6.00-7.00 sec 1.54 Gbytes 13.3 Gbits/sec 0 1.70 Mbytes
[ ] 7.00-8.00 sec 1.63 Gbytes 13.6 Gbits/sec 4 371 Kbytes
[ ] 8.00-9.00 sec 1.43 Gbytes 12.6 Gbits/sec 2 324 Kbytes
[ ] 9.00-10.00 sec 1.58 Gbytes 12.7 Gbits/sec 2 1.58 Mbytes
- - - - -
ID Interval Transfer Bitrate Retx Cong
[ ] 0.00-10.00 sec 10.4 Gbytes 12.4 Gbits/sec 39 sender
[ ] 0.00-10.01 sec 10.4 Gbytes 12.4 Gbits/sec 39 receiver
ipert Demo
root@kali:~/Documents#
```

Рис. 17: Проверка конфигурации с помощью iPerf3

```
iperf Done.  
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem  
root@mininet-vm:/home/mininet#
```

Рис. 18: Восстановление конфигурации h1

Интерактивные эксперименты



```
host: h1@mininet-vm
RX packets 970  bytes 241612 (241.6 KB)
RX errors 0  dropped 0  overruns 0  frame 0
TX packets 970  bytes 241612 (241.6 KB)
TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 10
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.97 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.368 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.131 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.067 ms

--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9191ms
rtt min/avg/max/mdev = 0.058/0.391/2.967/0.863 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 1
0ms reorder 25% 50%
root@mininet-vm:/home/mininet#
```

Рис. 19: Добавление правила

Интерактивные эксперименты

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=10.3 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=10.3 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=10.3 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=10.3 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=10.3 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=10.2 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=10.7 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19088ms
rtt min/avg/max/mdev = 0.061/9.462/10.721/3.138 ms
root@mininet-vm:/home/mininet#
```

Рис. 20: Проверка после добавления правила

0% потерь пакетов

Интерактивные эксперименты

```
root@h1:~# ping -c 3 10.0.0.2
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=10.7 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19088ms
rtt min/avg/max/mdev = 0.061/9.462/10.721/3.138 ms
root@mininet-virtual-machine:~# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=10.3 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=10.2 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.3 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=10.5 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=11.2 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=10.9 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19057ms
rtt min/avg/max/mdev = 0.073/10.128/11.161/2.316 ms
root@mininet-virtual-machine:~# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-virtual-machine:~# sudo tc qdisc add dev h1-eth0 root netem duplicate 50%
root@mininet-virtual-machine:~#
```

Рис. 21: Задание правила с дублированием

```
root@mininet-vn:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.37 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.492 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.226 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=1.29 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.063 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.069 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.069 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.079 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.067 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.099 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.099 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.068 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.101 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.102 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.074 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, +7 duplicates, 0% packet loss, time 19383ms
rtt min/avg/max/mdev = 0.058/0.260/3.373/0.655 ms
root@mininet-vn:/home/mininet#
```

Рис. 22: Проверка на наличие дубликатов

```
mininet@mininet-vm:~$ mkdir -p ~/work/lab_netem_i/expname  
mininet@mininet-vm:~$
```

Рис. 23: Создание каталога

```
mininet@mininet-vm:~$ mkdir -p ~/work/lab_netem_ii/simple-drop  
mininet@mininet-vm:~$ cd ~/work/lab_netem_ii/simple-drop  
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$
```

Рис. 24: Создание simple-drop

Воспроизведение экспериментов

```
#!/usr/bin/env python3
# lab_netem_ii.py

"""
Single experiment.
Output: ping.dat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
import time

def setupNet():
    """Create an empty network and add nodes to it."""
    net = Mininet(controller=Controller,
                  ifconfig=ifconfig)

    info( "Adding controller(s)" )
    net.addController( "c0" )

    info( "Adding host(s)" )
    h1 = net.addHost( "h1", ip="10.0.0.1" )
    h2 = net.addHost( "h2", ip="10.0.0.2" )

    info( "Adding switch(s)" )
    s1 = net.addSwitch( "s1" )

    info( "Creating links(s)" )
    net.addLink( h1, s1 )
    net.addLink( h2, s1 )

    info( "Starting network(s)" )
    net.start()

    info( "Set delay(s)" )
    h1.cmdPrint( "tc qdisc add dev h1-eth0 root netem loss\n\t= 10%" )
    h2.cmdPrint( "tc qdisc add dev h2-eth0 root netem loss\n\t= 10%" )

    time.sleep(10) # Wait 10 seconds

    info( "Ping(s)" )
    h1.cmdPrint( "ping -c 100, h2.IP(), 'i' srcp '10.0.0.1'\n\t= set '10.0.0.1, 0.0.0.0' i set -e 'n/time//g' -e\n\t'x/time//g' > ping.dat" )

    info( "Stopping network" )
    net.stop()

if __name__ == '__main__':
    setLogLevel( "info" )
    setupNet()
```

Рис. 25: Создание lab_netem_ii.py

В этих строках задается значение потери пакетов для интерфейса хоста:

```
h1.cmdPrint( 'tc qdisc add dev h1-eth0 root netem loss  
~ 10%'  
h2.cmdPrint( 'tc qdisc add dev h2-eth0 root netem loss  
~ 10%'
```


Воспроизведение экспериментов

```

root@kali: ~# cat 06_07.py
#!/usr/bin/env python
"""
Simple experiment
Output: ping-lat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.link import setLogLevel, info
import time

def setupNet():
    """Create an empty network and add nodes to it."""
    net = Mininet(controller=Controller, waitConnected=True)

    info( " *** Adding controller(s) *** ")
    net.addController('c0')

    info( " *** Adding hosts(s) *** ")
    h1 = net.addHost( 'h1', ip='19.0.0.1' )
    h2 = net.addHost( 'h2', ip='19.0.0.2' )

    info( " *** Adding switch(es) *** ")
    s1 = net.addSwitch( 's1' )

    info( " *** Creating links(s) *** ")
    net.addLink( h1, s1 )
    net.addLink( h2, s1 )

    info( " *** Starting network(s) *** ")
    net.start()

    info( " *** Set delay(s) *** ")
    h1.configDict( { 'delay': add_del_h1+end_h1+sync_delay } )
    h2.configDict( { 'delay': add_del_h2+end_h2+sync_delay } )

    time.sleep(5) # Wait 5 seconds

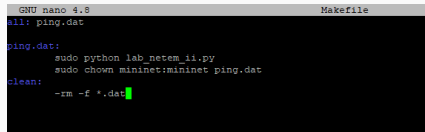
    info( " *** Ping(s) *** ")
    h1.configDict( { 'dstIP': '19.0.0.2' } )

    info( " *** Stopping network *** ")
    net.stop()

if __name__ == '__main__':
    setupNet()

```

Рис. 26: Коррекция скрипта



```
GNU nano 4.8                                Makefile
all: ping.dat

ping.dat:
    sudo python lab_netem_i1.py
    sudo chown mininet:mininet ping.dat
clean:
    -rm -f *.dat
```

Рис. 27: Создание Makefile

```
mininet@mininet-vm:~/work/lab_netem_11/simple-drop$ make
sudo python lab_netem_11.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem loss 10%')
*** h2 : ('tc qdisc add dev h2-eth0 root netem loss 10%')
*** Ping
*** h1 : ('ping -c 100 -W 1', '10.0.0.2', '| grep "packet loss" | awk \'{print $6, $7, $8}\'}')
100 packet loss,
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
```

Рис. 28: Запуск скрипта

Воспроизведение экспериментов

[illegible]

Рис. 29: Коррекция скрипта для ping.dat

```
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ ls  
lab netem ii.py Makefile ping.dat
```

Рис. 30: Наличие ping.dat

```
lab_netem_ii.py Maxville ping.dat  
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ cat ping.dat  
1013 1  
1.75 2  
0.263 3  
0.056 4  
0.077 5  
0.067 6  
0.080 7  
0.069 8  
0.083 9  
0.062 10  
0.061 11  
0.062 12  
0.062 13  
0.062 14  
0.064 15  
0.063 16  
0.077 17  
0.061 18  
0.072 19  
0.062 20  
0.062 21  
0.062 22  
0.062 23  
0.073 24  
0.049 25  
0.064 26  
0.064 27
```

Рис. 31: Содержимое ping.dat после повторного запуска скрипта

Задание для самостоятельной работы

Самостоятельно реализуйте воспроизводимые эксперименты по исследованию параметров сети, связанных с потерей, изменением порядка и повреждением пакетов при передаче данных

Задание для самостоятельной работы

```
#!/usr/bin/env python
from mininet.net import Mininet
from mininet.node import Controller
from mininet.log import setLogLevel, info
import time
import os

def cleanup():
    os.system('sudo ip link delete nl-eth0 2>/dev/null')
    os.system('sudo ip link delete nl-eth1 2>/dev/null')
    os.system('sudo ip link delete nl-eth2 2>/dev/null')
    os.system('sudo ip link delete nl-eth3 2>/dev/null')
    os.system('sudo rm -rf /var/lib/nl')

def setupNet():
    cleanup()

    net = Mininet(Controller, Controller, waitComponents=True)
    net.addController('c0')
    nl = net.addMininet('nl', ip='10.0.0.1')
    nl = net.addMininet('nl2', ip='10.0.0.2')
    nl = net.addMininet('nl3')
    net.addLink(nl, nl)
    net.addLink(nl, nl2)
    net.start()

    expressions = [
        ['txrate_0%', '1000 100%'],
        ['txrate_1%', '1000 100%'],
        ['txrate_2%', '1000 200%'],
        ['rxrate_0%', 'delay 10ms rxrate 100%'],
        ['rxrate_1%', 'delay 10ms rxrate 100%'],
        ['rxrate_2%', 'delay 10ms rxrate 100%'],
        ['rxrate_3%', 'rxrate 50%'],
        ['rxrate_4%', 'rxrate 100%'],
        ['rxrate_5%', 'delay 10ms rxrate 100%']
    ]

    for exp_name, exp in expressions:
        nl.cmd('tc qdisc del dev nl-eth0 root 2>/dev/null')
        nl.cmd('tc qdisc del dev nl-eth1 root 2>/dev/null')
        nl.cmd('tc qdisc add dev nl-eth0 root netem (no_red)')
        nl.cmd('tc qdisc add dev nl-eth1 root netem (no_red)')
        time.sleep(2)
        nl.cmd('tc qdisc -F -M 1 0:1 0:1 < state <exp_name> log 100')
        nl.cmd('tc qdisc <exp_name> state <exp_name> log 1 && <state 0% 0% 1% 2% 3% 4% 5% 6% 7% 8% 9% 10% 11% 12% 13% 14% 15% 16% 17% 18% 19% 20% 21% 22% 23% 24% 25% 26% 27% 28% 29% 30% 31% 32% 33% 34% 35% 36% 37% 38% 39% 40% 41% 42% 43% 44% 45% 46% 47% 48% 49% 50% 51% 52% 53% 54% 55% 56% 57% 58% 59% 60% 61% 62% 63% 64% 65% 66% 67% 68% 69% 70% 71% 72% 73% 74% 75% 76% 77% 78% 79% 80% 81% 82% 83% 84% 85% 86% 87% 88% 89% 90% 91% 92% 93% 94% 95% 96% 97% 98% 99% 100%')
        results = nl.cmd('cat /proc/net/route state <exp_name> log')
        info('<exp_name>: %s\n' % results)

    net.stop()

if __name__ == '__main__':
    setupNet()
    cleanup()
```

Рис. 32: Скрипт для самостоятельного задания

Задание для самостоятельной работы

```
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ make clean
rm -f *.dot
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ make
sudo python lab_netem_ii.py
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
loss_5: 50 packets transmitted, 37 received, 26% packet loss, time 50134ms
loss_10: 50 packets transmitted, 41 received, 18% packet loss, time 50182ms
loss_20: 50 packets transmitted, 34 received, 32% packet loss, time 50185ms
reorder_25: 50 packets transmitted, 50 received, 0% packet loss, time 49134ms
reorder_50: 50 packets transmitted, 50 received, 0% packet loss, time 49254ms
corrupt_5: 50 packets transmitted, 46 received, 8% packet loss, time 50165ms
corrupt_10: 50 packets transmitted, 41 received, 18% packet loss, time 50154ms
combined: 50 packets transmitted, 39 received, 22% packet loss, time 49230ms
*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
```

Рис. 33: Запуск обновленного скрипта

Задание для самостоятельной работы

[illegible]

Рис. 34: Пример одного из файлов воспроизводимого эксперимента

Выводы

В результате выполнения данной лабораторной работы я получила навыки работы проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.